

AI-Based Intrusion Detection System for IoT Networks Using Machine Learning and Deep Learning

Project Domain

- Cyber Security
 - Internet of Things (IoT)
 - Artificial Intelligence
 - Machine Learning
 - Deep Learning
 - Network Security
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Problem Statement

The rapid growth of Internet of Things (IoT) devices has significantly improved automation in domains such as smart homes, healthcare, transportation, agriculture, and industrial automation. However, the increasing number of interconnected devices has also introduced serious cybersecurity threats, including Denial-of-Service (DoS), Distributed Denial-of-Service (DDoS), botnet attacks, port scanning, spoofing, and other network intrusions.

Traditional signature-based intrusion detection systems are unable to detect previously unseen attacks effectively. Therefore, there is a need for intelligent intrusion detection systems capable of learning network behavior and automatically identifying anomalous traffic using Machine Learning and Deep Learning techniques.

Project Description

The objective of this project is to develop an AI-based Intrusion Detection System (IDS) capable of identifying malicious activities in IoT networks using network traffic data.

Students are expected to collect or use publicly available IoT intrusion datasets, preprocess the data, perform feature engineering, and develop an anomaly detection model using Machine Learning or Deep Learning algorithms.

The developed model should classify network traffic as either Normal or Malicious, and optionally identify the type of cyberattack.

The system should provide a user-friendly dashboard to visualize network traffic, detected attacks, model performance, and alert notifications.

Students are encouraged to compare multiple learning algorithms and evaluate their performance using standard cybersecurity metrics.

The methodology should include data preprocessing, feature selection, model training, validation, testing, and performance evaluation.

Objectives

- Develop an intelligent intrusion detection system for IoT environments.
 - Detect anomalous network traffic using AI techniques.
 - Classify network traffic into normal and malicious categories.
 - Identify different attack types.
 - Compare Machine Learning and Deep Learning models.
 - Visualize intrusion detection results through a dashboard.
 - Reduce false alarm rates while maintaining high detection accuracy.
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Functional Requirements

The proposed system should support:

1. Importing IoT network datasets
 2. Data preprocessing and normalization
 3. Feature selection or feature engineering
 4. Binary intrusion detection
 5. Multi-class attack classification (need to work once binary class is completed)
 6. Model training and testing
 7. Performance evaluation
 8. Attack visualization dashboard
 9. Alert generation
 10. Historical attack logs
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Suggested Datasets

Students may use one or more publicly available datasets such as:

- NSL-KDD
- CICIDS-2017
- IoTID20
- UNSW-NB15
- Bot-IoT
- ToN-IoT

Suggested Algorithms

Machine Learning

- Logistic Regression
- Decision Tree
- Random Forest
- K-Nearest Neighbors
- Support Vector Machine
- Naïve Bayes
- XGBoost

Deep Learning

- Artificial Neural Network (ANN)
- Convolutional Neural Network (CNN)
- LSTM
- GRU
- Autoencoder
- Transformer-based Models

Suggested Software Tools

- Python
- Scikit-learn
- TensorFlow / Keras
- PyTorch
- Pandas
- NumPy
- Matplotlib
- Streamlit / Flask
- Jupyter Notebook

Expected Deliverables

Students should submit:

1. Dataset preprocessing pipeline
 2. Trained AI model
 3. Intrusion Detection Dashboard
 4. Performance comparison report
 5. Source code
 6. Technical report
 7. Presentation
 8. Demonstration video
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Expected Output

The completed system should be able to:

- Detect anomalous IoT network traffic.
 - Classify attacks in real time or from recorded traffic.
 - Display attack statistics and alerts.
 - Compare the performance of different AI algorithms.
 - Generate confusion matrices and performance reports.
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Possible Enhancements

Students should implement one or more advanced features:

- Real-time packet capture using Wireshark/Scapy
- Live network traffic monitoring
- Explainable AI (SHAP/LIME)
- Federated Learning
- Edge AI deployment on Raspberry Pi
- Incremental learning for new attacks
- Threat intelligence integration
- SIEM dashboard integration
- Cloud deployment
- Email/SMS alerts

- WebSocket-based real-time visualization
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Learning Outcomes

After completing this project, students will be able to:

- Understand IoT network security challenges.
 - Perform cybersecurity data preprocessing.
 - Develop AI-based intrusion detection models.
 - Compare machine learning and deep learning techniques.
 - Evaluate cybersecurity models using standard performance metrics.
 - Design secure and intelligent IoT applications.
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Recommended Technology Stack

Component	Suggested Technology
Programming Language	Python
Data Processing	Pandas, NumPy
Machine Learning	Scikit-learn
Deep Learning	TensorFlow / PyTorch
Visualization	Matplotlib, Plotly
Dashboard	Streamlit / Flask
Database (Optional)	MongoDB / MySQL
Deployment	Docker / Raspberry Pi / Cloud

Notes for Students

The solution should be capable of handling binary intrusion detection (Normal vs. Attack) and, if possible, multi-class attack classification (e.g., DoS, DDoS, Botnet, Port Scan, Spoofing).

Students are encouraged to compare at least three Machine Learning models and one Deep Learning model, analyzing metrics such as Accuracy, Precision, Recall, F1-Score, ROC-AUC, and False Alarm Rate before selecting the final model.